CS 301 · MEETING 13 · RECAP 1

Lines, Planes, and a Walk

Linear systems, their geometric solutions, sensitivity to changes in the data, and a random walk on a graph.

A linear system in matrix form

$$2x + y = 5, \quad x - y = 1 \qquad \Longrightarrow \qquad A = \begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix},\ x = \begin{bmatrix} x \\ y \end{bmatrix},\ b = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$$

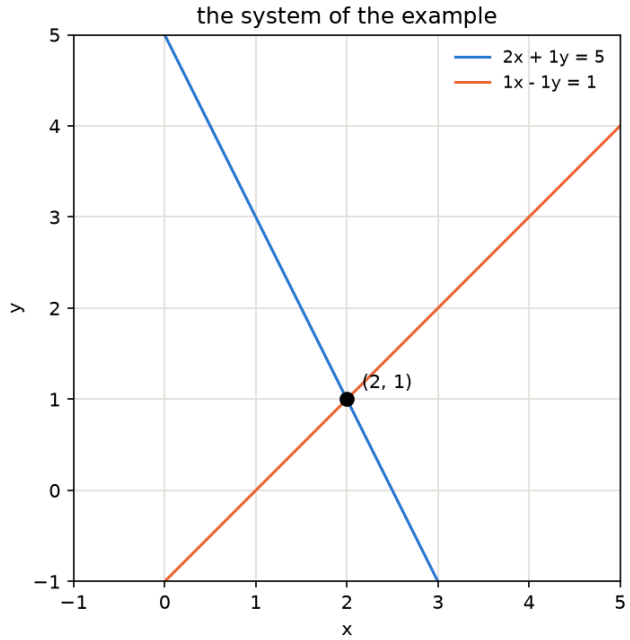
- **Linear**: a sum of unknowns multiplied by coefficients
- One row per equation; `A @ x` evaluates their left-hand sides
- `np.linalg.solve(A, b)`: the x that makes `A @ x == b`

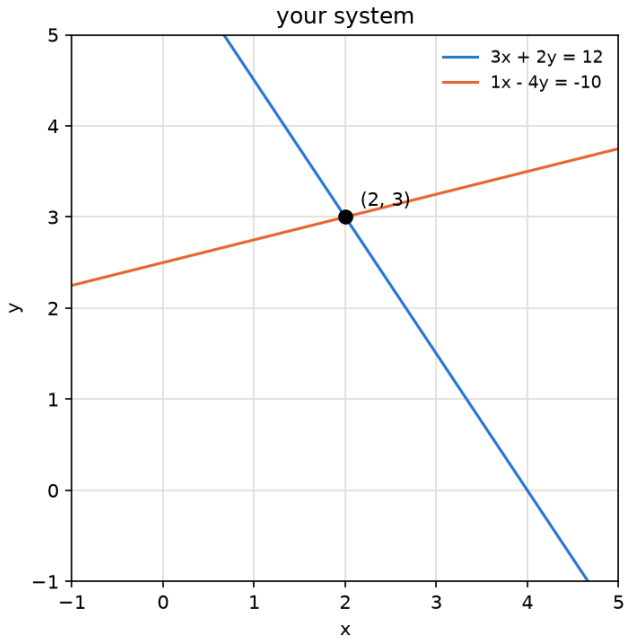
Elimination, by hand

$$2x + y = 5 \qquad x - y = 1 \qquad \text{add: } 3x = 6 \;\Rightarrow\; x = 2,\; y = 1$$

- Use one equation to remove an unknown from the other
- Substitute the value back to find the other unknown
- `solve` uses elimination in compiled code

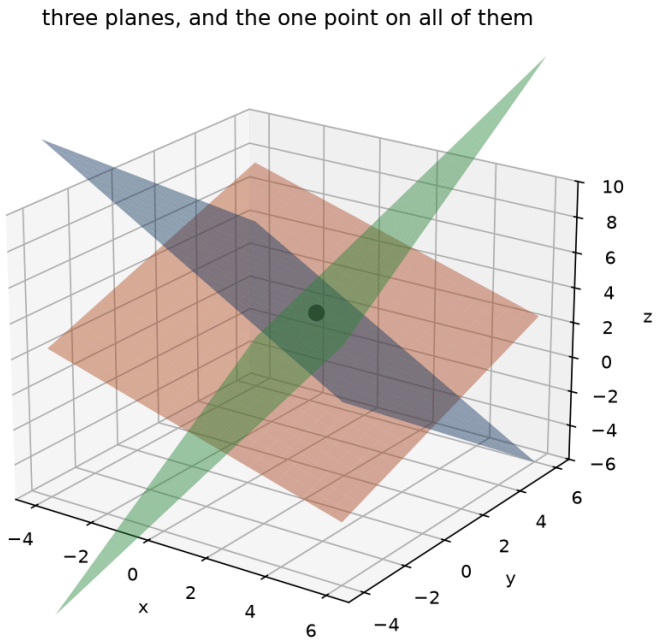
An equation is a line; the solution is where lines cross





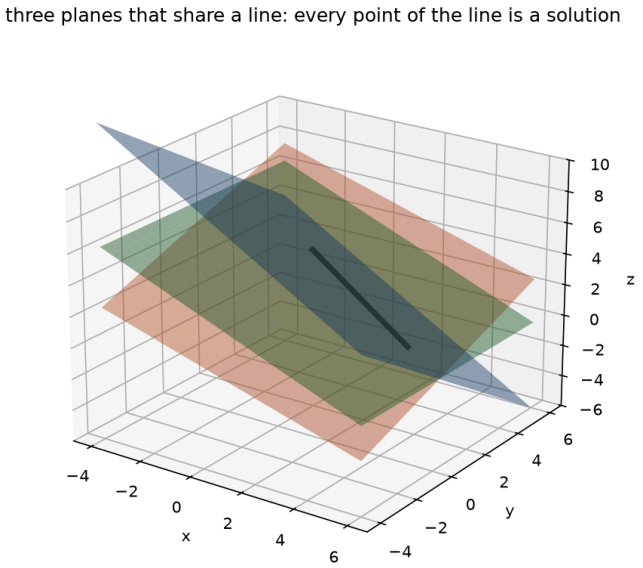
- All the points (x, y) that satisfy one equation: a line
- A solution satisfies both: a point on both lines

Three unknowns: three planes, one point



- An equation in three unknowns: a plane
- Here, two planes meet along a line; the third selects one point

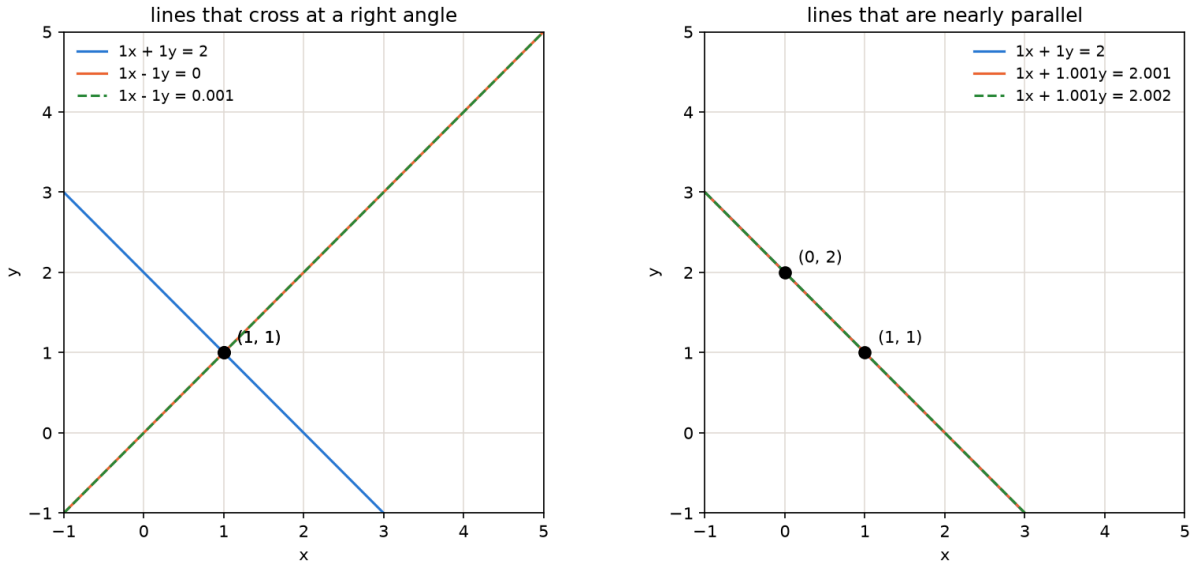
When the third plane adds nothing




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NumPy refuses the system: Singular matrix
Correct: the three equations hold only 2 independent ones, and the planes share a line.
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- The third equation = the first + the second: **rank** 2, not 3
- Infinitely many solutions; with another right-hand side, none

Nearly parallel: the condition number



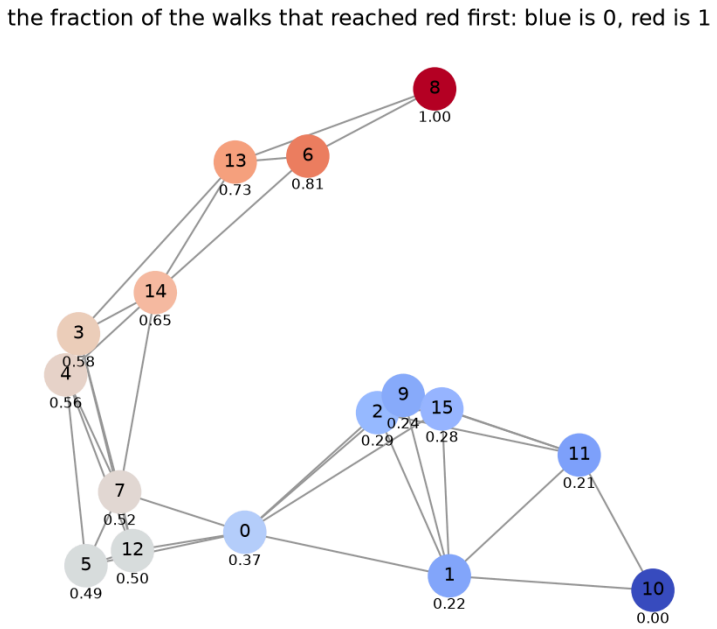

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lines that cross at a right angle:
   the solution [1. 1.], after the change [1.0005 0.9995]: moved by 0.0007
   condition number 1
lines that are nearly parallel:
   the solution [1. 1.], after the change [0. 2.]: moved by 1.4142
   condition number 4,002
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Conditioning and reliable digits

condition number	relative-error amplification	digits retained, worst-case estimate
1	at most 1×	16
10^4	up to 10,000 times	about 12
10^8	up to 10^8 times	about 8
10^{12}	up to 10^{12} times	about 4
10^{16}	up to 10^{16} times	none

- Worst-case estimate: potential digits ≈ $\max(0, 16 - \log_{10}(\text{condition number}))$
- Six-digit data, condition number 10^6: no reliable digits guaranteed
- Use `np.linalg.cond(A)` to assess sensitivity

A walk on a graph



- From each node: a random neighbour, again and again, until red or blue
- 2,000 walks from each node: the fraction that reached red first

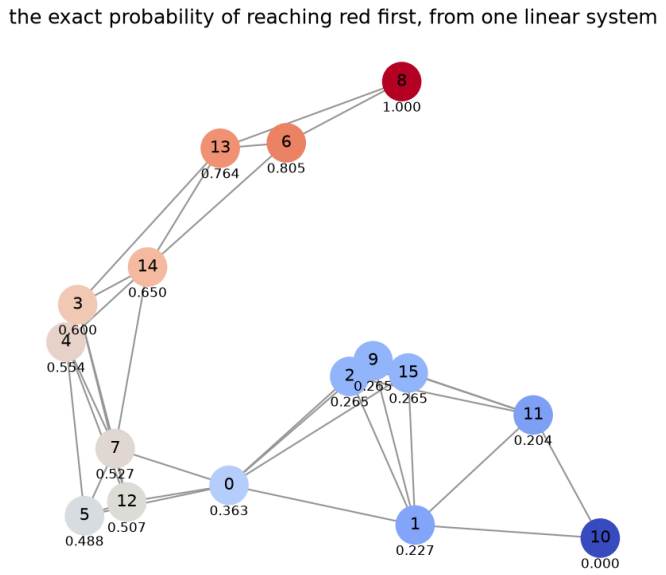
The equations for the walk

$$x_v = \frac{1}{\deg(v)} \sum_{u \sim v} x_u \quad \text{for every other node,} \qquad x_{\text{red}} = 1,\ x_{\text{blue}} = 0$$

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node 6 has the neighbours [8, 13, 14]: its equation is x6 = (x8 + x13 + x14) / 3
the row of node 6 in the matrix: 1 at column 6, -0.333 at column 8, -0.333 at column 13, -0.333 at
column 14, 0 elsewhere
```

- One probability per node: 16 unknowns and 16 equations
- deg(v), the **degree** of v: its number of neighbours
- Average the neighbours' probabilities after the first step

Direct solution and 2,000 simulated walks




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largest difference between the 2,000 walks and the exact solution: 0.032

 node  neighbours  simulated   exact
    0           7       0.37   0.363
    1           6       0.22   0.227
    2           5       0.29   0.265
    3           5       0.58   0.600
```

- The simulation approximates the probabilities found by `solve`

From two unknowns to a million

- Matrix form applies to systems with 2 or 1,000,000 unknowns
- Large systems require appropriate storage and solvers
- Recap 2: operation counts, round-off, and sparse matrices